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A Holistic Framework for Evolving AI-based Systems

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***Abstract*—Building easily *modifiable* software systems is a challenge for engineers. *AI-based systems* add AI models and data, which change frequently. This research identifies *change scenarios* and their impact in a *framework* for modifiable AI-based systems. This holistic framework consists of three architectural views and three AI-based themes. We apply this framework to various *industry cases*, to elicit which factors drive the evolution.**

***Index Terms*—AI engineering, modifiability, enterprise architecture**

I. Introduction

Most of the cost of developing software systems occurs after their initial release, thus challenging software engineers to embrace change. The continual change that software undergoes during its lifetime is generally called *evolution*, and the degree to which it is easy or hard to change existing software is often called *modifiability* [1]. Other quality criteria are closely related: modularity, reusability, analyseability, testability, flexibility, adaptability, scalability, installability, and replaceability [2]. In our research, we use the broader term modifiability to refer to all these aspects of evolving IT systems.

Today, the industry is able to quickly integrate existing AI components into their software architecture. These software systems are generally referred to as AI-based systems. The need for AI-based systems that are reliable and aligned with human expectations has been underlined by two publications from the European Commission (European Ethics Guidelines for Trustworthy AI, 2019[1] and the European AI Act, 2024[2]).

Reliable AI components introduce changes due to the reliance on a machine learning model and (training) data [3], [4]. An example is that new training data from monitoring the system in production is gathered, and used to retrain the model, causing the model to change more often than the code itself. Another example is that updating the model with a better version often changes the model architecture, feature selection, or hyper-parameters.

That an AI-based system needs to be *aligned to human expectations* introduces new functionalities in the user interface, monitoring dashboards or system logging components. Some examples are: how can effective validation of the functional suitability of the system be achieved? Or, how to handle data labeling, or how should we validate the predictions of the model? And finally, how can we enable those users, who have the experience and ability, to validate the accuracy of automated tasks?

The modifiability of AI-based systems is not only happening in source code but also in AI models and the data that the model is using. In the proposed research, we take these three themes into account in the proposed framework.

Although many authors have identified challenges and best practices for evolving AI-based systems [3]–[22], specific modifiability mechanisms for reliable AI-based systems aligned to human expectations, remain scattered, incomplete, and under-validated. Our goal is to address this gap.

II. Research Approach

We will build a framework for modifiability, based on existing literature and industry field studies. This holistic framework consists of two parts. The first part is the identification of typical change scenarios for AI-based systems. We aim to infer these scenarios empirically from industrial cases. We will link these scenarios to maturity levels and strategic roadmaps. The second part is a holistic perspective of the systems' development. This holistic perspective consists of 3 views and 3 AI-based themes. The three views are the *Organization and Teams* view, the *Tools, Software and Infrastructure* view, and thirdly, the *Processes* view. The three AI-based themes are *Data*, *AI Model* and *Software*. One contribution that we envision for our framework is to identify and model the dependencies between these views and themes and particular change scenarios.

While building this framework, we will address the following research questions.

A. Research Questions

The central research question of this PhD is:

How to support the evolution of AI-based systems with a holistic framework linking change scenarios to architectural perspectives and AI themes?

The research questions to answer are

- (**RQ1**) *What are typical change scenarios for AI-based systems?*
- (**RQ2**) *How do design decisions/patterns facilitate or inhibit change scenarios?*

[1]https://ec.europa.eu/digital-single-market/en/high-level-expert-group-artificial-intelligence

[2]https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai

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- (**RQ3**) *What are the best practices in engineering AI-based systems?*

B. Research Methods

As noted by [10], [11], case study research has emerged as the predominant research method for investigating AI-based systems in software engineering. One of our aims is to examine mature industrial-scale applications, while encompassing the entire process workflow for AI-based systems, with particular emphasis on modifiability, and collect best practices.

We will apply an iterative methodology within the field of software engineering to examine contemporary phenomena pertinent to software process improvement, specifically introduced by maintaining reliable AI-based systems, aligned to human expectations.

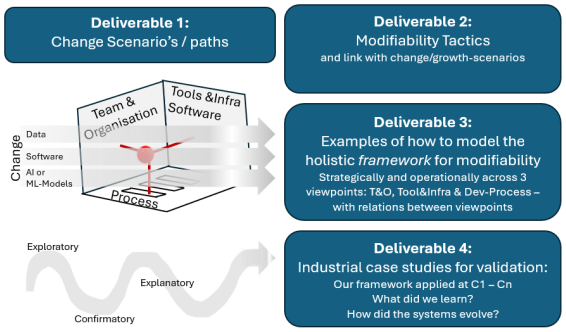

Fig. 1. We will iteratively observe industry cases to deliver a holistic framework with 3 views and 3 AI-based themes.

The spiral research process, as elaborated upon by [23], is designed to be executed iteratively with ongoing feedback. We will apply this process to the approximately 40 AI-enabled system projects that are executed annually by our students. We have ongoing contacts with industrial partners in various domains (e.g., marketing, hospital, manufacturing) with mature AI-enabled systems. In these companies, IT architects and AI- or software engineers will be interviewed to identify changes in their AI-based systems: both on a strategic (long term) and an operational (short term) level.

III. Contributions

Our research, as outlined in Figure 1, will produce four deliverables, each discussed below.

A. Collection of Change Scenarios (RQ1)

Through structured interviews, surveys and literature study we collect, analyse and cluster change scenarios specific to AI-based systems. We identify typical evolution paths (i.e. a sequence of change scenarios) which enable incremental automation, scaling-up and maturity.

B. Impact on Views and Themes (RQ1, RQ2)

Within each of our views, we consider three common themes in the development of AI-based systems: Data Engineering, Software Engineering and Model Engineering. For each change scenario, we aim to elicit the impact on these themes. We identify the impact of change scenarios for data, software and AI models in three views, namely 1) Teams and Organization, 2) Tools, Infra and Software and 3) Processes. A visualization of this result can be found in Figure 2.

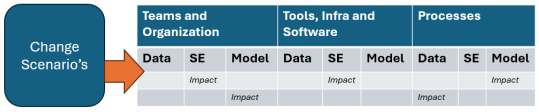

Fig. 2. Impact of change for AI-based systems mapped onto 3 views and themes.

This framework and a collection of best practices will be shared with practitioners.

C. Modeling our Framework (RQ1, RQ2, RQ3)

We use the framework to model specific case studies. This illustrates how modeling methodologies (such as e.g. TOGAF) can be used in support of our method. Aligning the tools to the change scenarios, can be done using reference architectures. We will focus on *classical, facilitation architectures* which are based on existing experiences in research and industry and aim to introduce a positive impact on the AI-engineering domain [24].

D. Applying the Framework to Mature AI-based Systems (RQ3)

We will apply our framework to a number of industrial case studies. In these case studies, we will observe the relationships and the time dimension as described in [25]. One of our industry partners is, for example, focusing on extending their architecture to realize monthly changes in production lines for complex components and automatic visual anomaly detection [26]. Through these case studies, we learn which modifiability scenarios are used, which design patterns/decisions are made and how they cut across the various views.

E. Validation

Validation of our approach consists of applying the framework with a variety of industrial case studies. The insights from these case studies will be used to refine the framework. Also, this should elicit which contextual factors play a role in driving the design and evolution of AI-based systems. We apply various techniques (like independent coding) for increasing the internal validity.

IV. Conclusion

A major concern in the development of software systems is their ability to evolve. This concern is amplified in trustworthy AI-based systems. We propose to develop a framework that aids in the design of AI-based systems from a modifiability perspective. We will develop this framework empirically based on industrial case studies.

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